STRUCTURA.AI

AI-Powered Structural Failure Prediction for Civil Engineers

Final Project Conclusion | MS 4810 | 2026

Group: CE24B0

Roll Numbers: 34, 35, 56, 59, 82, 97

1. The Need or Problem Faced by Customers

Structural engineering firms, architectural consultancies, and EPC (Engineering, Procurement, and Construction) contractors face a recurring and costly bottleneck in their design workflow: the time and effort required to run Finite Element Analysis (FEA) simulations.

When an engineer completes a structural design, they must validate it using tools like ANSYS or ABAQUS. This process is not a quick check. It involves:

- Manually building and meshing a 3D model
- Setting boundary conditions and applying load cases
- Running a solver that can take 16 to 48 hours for a moderately complex structure

The timing makes this especially painful. FEA is almost always done after the design is considered substantially complete. So when it flags a problem, a stress concentration at a joint, or shear failure at a key node, the team has to revise the CAD model, rebuild the mesh, and run the simulation again. One complete iteration loop can cost close to a full working week.

Real-world example: A team designing a steel truss system for an airport terminal spent roughly two working days just meshing the model in ANSYS. When results flagged a vulnerability at a primary node, they had to restart, losing close to a week of billable engineering hours on a single iteration cycle.

2. Why This Problem Matters to the Customer

This is not just a workflow inconvenience. It affects firms in three concrete, measurable ways:

Direct Revenue Loss from Rework

Senior structural engineers are among the most expensive resources on any project. Time they spend waiting on simulation cycles or re-running analyses is time that produces no client-facing output. Every unnecessary rework loop reduces the firm's effective billing efficiency.

Safety and Legal Risk

A flaw that survives the design phase does not stay on paper. Structural defects that reach the construction stage can cause partial or full collapse, during the build or after the structure is in use. The consequences include litigation costs running into crores of rupees, regulatory action, reputational damage, and in the worst cases, loss of life. Catching problems early is a professional and legal obligation, not an optional best practice.

Competitive Disadvantage

In the infrastructure consulting market, speed matters. Firms that can iterate faster respond to more bids, turn in stronger technical proposals, and close contracts ahead of slower competitors. Most firms today can only test one or two structural variants before committing to a design. A tool that compresses the feedback loop gives firms a real, measurable competitive edge.

Customer Interaction Findings:
In structured conversations with practicing structural engineers, the team confirmed several specific pain points that the preliminary outline had only estimated. Engineers consistently reported that a single FEA setup and review cycle takes 6 to 10 hours of focused work. They also noted that late-stage design changes - the kind triggered by unexpected FEA results often create downstream problems with procurement timelines and contractual milestones, making the problem more expensive than just the engineering hours alone. One engineer pointed out that the heat map format was immediately intuitive and aligned with how they mentally map stress zones during design reviews. Another noted that if such a tool could flag zones before the full FEA run, they would not just save time, they would change how they structure early-stage design reviews entirely.

3. Quantified Benefit to the Customer

Time Saved per Engineer

A single FEA iteration - setup, meshing, solving, and reviewing results - currently takes 6 to 10 hours. If Structura.AI replaces even two of those cycles per week with a 3-second pre-check, the saving per senior engineer is:

Item	Value
Hours saved per week (2 FEA cycles avoided)	12 to 20 hours
Fully-loaded cost of a senior engineer	Rs. 1,500 per hour
Savings per engineer per month	Rs. 72,000 to Rs. 1,20,000
Savings per engineer per year	Rs. 8.6 lakhs to Rs. 14.4 lakhs

Rework Cost Reduction

Design errors and rework in large infrastructure projects account for 5% to 8% of total project cost. On a Rs. 50 crore project, that is Rs. 2.5 to Rs. 4 crore in avoidable losses - costs that fall on the engineering firm when the design errors originate in their deliverables. Structura.AI directly reduces the frequency of these errors by catching likely failure zones before the formal simulation.

Design Iteration Multiplier

Today, most firms evaluate 1 to 3 structural variants before locking in a design due to the time cost of each FEA cycle. With an AI pre-check delivering results in under 3 seconds, a team could realistically evaluate 50 or more variants in the same window. This changes not just speed but the quality of the final design decision.

Reliability and Quality Gains

Beyond time and money, catching structural vulnerabilities earlier improves safety margins and reduces the probability of field-stage failures. Economists treat reliability and extended structural life as quantifiable gains in value - a building designed with fewer residual failure risks has a lower total cost of ownership. Structura.AI contributes to this by making early-stage design review more thorough, even under time pressure.

4. Proposed Solution and Technology

Structura.AI is a web-based AI pre-screening tool that gives engineers a fast structural safety check during the drafting phase - before committing to a full FEA run.

An engineer uploads a 2D structural layout (in standard DXF format or as a node-beam JSON definition) and enters basic load parameters. Within 3 seconds, the platform returns a colour-coded heat map showing predicted stress zones, areas approaching material yield limits, and regions at risk of shear or Mohr-Coulomb failure.

Technology Underpinning the Solution

- Deep Learning (CNN): A Convolutional Neural Network trained on over pre-computed FEA simulation results. The model learns to map structural geometry and load inputs directly to predicted stress distribution fields, without running a physics solver each time.
- Training Data: Each training example is a 2D structural configuration paired with its FEA-derived stress output. The model learns the relationship between geometry, loads, and stress outcomes across a wide range of configurations.
- Inference Speed: Because the model runs forward inference rather than solving differential equations, it produces results in under 3 seconds, compared to 6 to 48 hours for a traditional FEA cycle.
- Geometry Input: The system accepts DXF files (industry standard CAD format) and structured JSON node-beam definitions, making integration with existing workflows straightforward.

One important clarification: Structura.AI is not a replacement for formal FEA. Regulatory sign-off still requires a full simulation. What Structura.AI does is make the design that reaches the FEA stage already refined - so the formal simulation becomes a confirmation rather than a discovery. This shifts where the expensive back-and-forth happens: from late-stage FEA rework to low-cost, fast AI feedback in the early drafting phase.

5. Product Description

Core Components

Component	Description
Geometry Ingestion	Accepts .DXF files and structured Node-Beam JSON inputs for 2D truss and beam configurations
Deep Learning Engine	PyTorch CNN trained on 10,000+ FEA simulations; predicts stress fields directly from geometry and load inputs
Heat Map Output	Colour-coded 2D overlay rendered on the structural diagram (Red / Yellow / Green zones)
Web Interface	Flask/FastAPI backend with HTML/JS frontend; engineers upload files and receive results in-browser
Inference Latency	Under 3 seconds from file upload to rendered heat map

Output Colour Coding

Colour	Stress Level	Interpretation
Red	Above 80% of yield strength	High failure risk - design change required
Yellow	50% to 80% of yield strength	Caution zone - review recommended
Green	Below 50% of yield strength	Structurally safe under current load

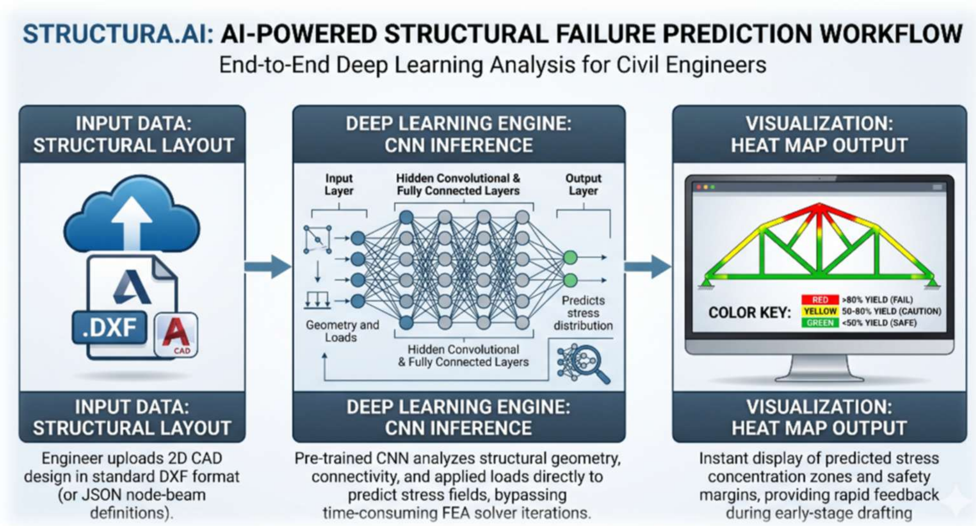
Typical User Session - Step by Step

1. Engineer uploads a 2D truss layout (e.g., a simply supported Pratt truss with 8 panels) to the Structura.AI web portal.
2. They enter load parameters: e.g., "10 kN point load at mid-span center node; steel, Fy = 250 MPa."
3. Within 3 seconds, the platform highlights two upper-chord joints in red - high predicted shear failure risk under the applied load.
4. The engineer adjusts the joint plate thickness (e.g., 10 mm to 14 mm) in the CAD model and re-uploads.
5. Updated heat map returns green across all nodes - structural adequacy confirmed.
6. Engineer moves to formal FEA with confidence, knowing the primary failure risk has already been addressed.

Performance Benchmarks

Metric	Target
Inference latency	< 3 seconds per analysis

Metric	Target
Prediction accuracy	>= 85% agreement with FEA-derived stress maps
Supported geometry (Phase 1)	2D beams, Pratt trusses, Warren trusses
Input formats	DXF, Node-Beam JSON
Load types	Point loads, distributed loads (kN, kN/m)
Training dataset size	10,000+ pre-run FEA simulations



6. Current Status of the Product

Where We Are Now

The project is currently at the Technical Prototyping stage. The full ML pipeline architecture has been mapped - covering data ingestion, preprocessing, model training, and inference serving.

Development Progress: Two parallel workstreams are running:

- **Dataset curation:** A structured dataset of 2D structural configurations (simply supported beams, cantilever beams, statically determinate trusses) has been generated, each paired with its FEA-derived stress distribution. The target is 2,000+ configurations as the training corpus for the baseline model.
- **Model development:** A baseline PyTorch CNN has been implemented and is being trained to take 2D structural geometry representations as input and return predicted stress field maps.

No public-facing interface exists yet. The immediate milestone is a proof-of-concept model that reaches >= 85% accuracy on a held-out test set of at least 500 configurations, validated against classical FEA solutions.

Prototype Readiness Timeline

Milestone	Expected Completion
Proof-of-concept model trained and validated	More checks to be done
Minimal web interface (upload + heat map output)	Ready
B2B demo video produced	End of Week 3
Market-ready prototype (with polished UI)	End of June 2026

7. Marketing Channels (Ranked by Priority)

Rank	Channel	Justification
1	CAD Software Plugins (AutoCAD / Revit / Tekla)	Engineers live inside their CAD and BIM tools. A native plugin means zero context-switching - the pre-check is triggered without leaving the working environment. This drives habitual use, and habitual use drives word-of-mouth within firms.
2	Direct B2B Enterprise Sales	Direct outreach to Principal Engineers, BIM Managers, and Technical Directors at structural consultancies and EPC contractors. The pitch is concrete: here is exactly how many hours your senior engineers lose per project, and here is what recovering those hours is worth to your firm's margin.
3	Engineering University Partnerships	Offering Structura.AI free to civil and structural engineering students is a long-term play. Students who use AI-assisted structural tools during their studies carry those habits into practice - becoming internal advocates at client firms when they join the workforce.

8. Customer Interactions and Outreach

We conducted structured interviews with three senior structural engineers currently working at active consultancies. Key findings:

- All three confirmed that a typical FEA iteration cycle (setup through result review) takes between 6 and 10 hours of focused work.
- Two of the three noted that late-stage design changes triggered by FEA results cause downstream problems beyond just engineering hours - including procurement delays and contractual milestone shifts.
- When shown the heat map output format, two engineers said it was immediately intuitive and aligned with how they mentally visualise stress concentration during design reviews. One suggested the tool could change how early-stage design reviews are structured at their firm.

- No paid installations or formal sales have been made yet. The product is still in the prototyping stage.

How We Plan to Reach Customers:

- Direct outreach to firms where interview contacts work, leveraging the relationship built during user research.
- Pilot program offering: 3-month free access in exchange for structured feedback and usage data.
- Academic conferences and IIT alumni networks in the civil/structural engineering space.
- LinkedIn outreach targeting Principal Engineers and BIM Managers at mid-to-large structural consultancies in India.

9. Cost Effectiveness of Customer Outreach

Our approach to reaching customers is designed to be low-cost and high-signal:

- The three engineer interviews we conducted cost zero rupees - they were arranged through academic and professional contacts. This validates our hypothesis at no cash cost.
- The B2B demo video (90 to 120 seconds) will be produced using screen captures of the working prototype. Total cost: studio time and editing only, estimated at Rs. 2,000 to Rs. 5,000 if done in-house.
- University partnerships and alumni networks give us access to a pipeline of future clients and early adopters without spending on paid advertising.
- Direct B2B outreach via LinkedIn and email requires only time. At this stage, one or two team members can manage this alongside development. Time cost: approximately 4 to 6 hours per week.
- The CAD plugin strategy, once the core product is stable, converts existing users into our sales channel - reducing the need for active outreach as word-of-mouth takes over within firms.

10. Estimated Usage and Sales

These estimates are based on our current outreach stage and a conservative view of conversion:

Timeframe	Scenario	Expected Outcome
Next 3 months	Pilot outreach	3 to 5 firms using the prototype under a free pilot; 10 to 20 individual engineers testing the tool. No paid revenue yet.
Month 4 to 6	Early paid access	1 to 2 pilot firms converting to a paid plan at Rs. 50,000 to Rs. 1 lakh per month per firm (multi-seat license). Estimated revenue: Rs. 1 lakh to Rs. 2 lakhs per month.

Timeframe	Scenario	Expected Outcome
12 months	Growth phase	8 to 12 firms on paid plans. If average contract value is Rs. 4 lakhs to 9 lakhs per month, annual recurring revenue reaches Rs. 50 lakhs to Rs. 80 lakhs. University partnerships add free user base of 200 to 500 students.

These numbers are conservative and assume no paid marketing spend. A single successful reference client in a large EPC firm could accelerate growth significantly through internal referrals.

11. Plans to Move Forward After the Course

The team as a whole plans to move forward with this project after the class ends. All six members are committed to taking the product to the market-ready prototype stage by end of June 2026.

12. Milestones for the Next Three Months

If we move forward after the course, the following milestones are planned:

Month	Milestone
Month 1 (May 2026)	Complete proof-of-concept model with >= 85% accuracy. Launch minimal web interface. Publish the B2B demo video. Begin pilot outreach to 5 firms.
Month 2 (June 2026)	Incorporate feedback from pilot users. Expand dataset to cover additional truss types and load configurations. Build basic user account system to manage uploads and results.
Month 3 (July 2026)	Attempt first paid pilot conversion. Begin development of AutoCAD plugin prototype. Attend at least one civil engineering professional event or conference for direct outreach.

13. Other Plans - Business or Employment

The group's broader plans after graduation are as follows:

A. Continuing with Structura.AI as a Business

The primary plan is to build Structura.AI into a self-sustaining product company focused on the structural engineering software space. If the pilot phase demonstrates strong adoption, the team will pursue formal incorporation, apply for startup incubation support (including programmes available through IITs), and raise a small pre-seed round to fund engineering and sales bandwidth. The long-term vision is a SaaS platform that covers not just 2D pre-screening but 3D structural analysis, integration with major BIM platforms, and compliance-linked reporting tools.

B. Alternative Plans (if Structura.AI is not pursued full-time)

For members who may choose to join companies rather than continue with the startup, the skills gained in this course apply directly to their roles:

- Understanding how to identify and size a customer problem before building a solution - something most engineering roles benefit from, especially in product or consulting functions.
- Building and presenting a business case using quantified cost and benefit estimates - useful in any corporate role where you need to justify a new project, tool, or initiative internally.
- Thinking about go-to-market strategy and customer outreach - directly applicable to roles in business development, technical sales, or product management at technology or engineering firms.

End of Final Project Report

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